
Broadband Expansion and Small-Firm Productivity: Evidence from a Staggered Municipal Rollout

Referee Report

July 14, 2026

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Referee Report

Overall assessment

The paper studies an important and recognizable question with a useful small-firm setting, but its central causal interpretation is not supported by the rollout rule or the displayed event study. The staggered-adoption estimand and core inference remain unresolved; the model-to-regression bridge, mechanism interpretation, policy calculation, and priority claim also exceed the evidence; and an impossible employee summary statistic raises a basic data-reporting concern. These are root problems for the paper's contribution at a leading field journal, not matters of presentation alone.

The main strengths worth preserving are:

- The paper studies a recognizable broadband-and-firm-productivity question and combines small-firm administrative data with a task-intensity heterogeneity design.
- The manuscript reports the treatment-by-2008-communication-intensity coefficient transparently as 0.032 with a standard error of 0.005.

Recommendation and main grounds

Recommendation: Reject

At a leading field journal, the current paper does not clear the identification and internal-validity bar: the policy schedule explicitly selects on projected growth, the event-study leads visibly differ from zero, the staggered TWFE comparison is not characterized, and several independent model, policy, novelty, and data-reporting claims are unsupported or internally inconsistent.

The assessment would improve with these revisions:

- Recover a clearly defined causal estimand with a design that directly addresses the rollout score and displayed pre-trends, uses appropriate staggered-adoption comparisons and municipality-level inference throughout, and shows that post-treatment sample selection does not drive the result; otherwise narrow the paper to a descriptive contribution.
- Correct the production-model algebra and proposition, then either measure evidence that distinguishes task outsourcing from other communication-intensive channels or narrow the mechanism claim to heterogeneous association.
- Rebuild the policy and contribution claims from traceable quantities and verified literature, and resolve the impossible employee statistic before asking readers to rely on the empirical sample.

Issues that could prevent publication

1. The central causal design does not recover a credible or clearly defined effect

The rollout score selects municipalities partly on projected growth, while Figure 2 displays negative pre-treatment coefficients moving toward zero and post-treatment coefficients that continue rising. Standard TWFE under staggered adoption is left as an unspecified mixture, municipality clustering is not carried through the dynamic and heterogeneity results, and the balanced-survivor sample conditions inclusion on post-treatment outcomes. Together these features prevent a causal-average or permanent-effect interpretation of the reported 0.034 coefficient.

This requires additional analysis using the current evidence base.

What would change the assessment: Report cohort-time estimates against not-yet-treated or never-treated controls, condition the design on the published rollout score and engineering variables, carry municipality-level uncertainty through all headline analyses, and show robustness to sample rules that do not require post-treatment survival. If differential leads persist, remove the causal-average and permanent-effect claims.

2. The formal model does not support the estimating equation or the claimed mechanism

The derivation moves log labor to the left with the wrong coefficient and introduces a standalone broadband term not generated by the displayed linearization. The proposition is stated globally although its proof requires the switching regime, and the communication-intensity interaction cannot distinguish task outsourcing from other communication-related channels.

This can be corrected within the current design.

What would change the assessment: Correct the algebra and state the proposition only on the regime for which it is true. Then align the estimating equation with the model and either provide an outcome or design that distinguishes task outsourcing from competing communication channels, or present the interaction as non-structural heterogeneity.

3. The policy, contribution, and core data-reporting claims require rebuilding

The policy section converts a labor-productivity level coefficient into TFP, annual growth, aggregate euros, and fiscal repayment without a traceable bridge. The paper's first-evidence claim is contradicted by close pre-existing firm-level studies, and Table 1 reports an employee mean above its maximum. These errors affect the claimed endpoint, contribution, and credibility of the core sample description.

This can be corrected within the current design.

What would change the assessment: Rebuild the policy calculation from clearly defined and traceable objects or remove it, position the paper against the closest verified firm-level broadband studies, and reconcile Table 1 with the underlying data and every dependent result.

Other major issues

No other major substantive issues were identified.

Is the argument convincing?

The argument is not currently convincing. Its headline estimate is not separated from forecast-based rollout selection or the displayed differential leads; the formal mechanism is not valid as written; the policy endpoint is built from unsupported conversions; and the sample description contains an arithmetically impossible employee statistic.

Closest literature and key differences

The manuscript's relevant claim is: This is the first causal evidence on broadband expansion and small-firm productivity. That claim is not convincing as written. Prior firm studies already use broadband infrastructure variation to study firm outcomes, and the staged rural municipal-rollout study is a particularly close antecedent. A more defensible formulation is: "Relative to prior firm studies, this paper contributes evidence from a nationwide universal rollout for firms with fewer than fifty employees, with labor productivity as the outcome and communication intensity as the main heterogeneity margin."

Giulia Canzian, Samuele Poy, and Simone Schuller (2019), Broadband upgrade and firm performance in rural areas: Quasi-experimental evidence. The study uses a staged rural municipal broadband upgrade and firm balance sheets in a difference-in-differences design; it reports positive effects on revenue and total factor productivity. Infrastructure timing, municipality assignment, firm panels, productivity, and a causal DiD design overlap directly. The manuscript combines nationwide coverage, an under-fifty sample, labor productivity, and communication-intensity heterogeneity.

Suggested revision: Replace the first-evidence claim and compare treatment timing, sample, outcome, and surviving heterogeneity directly.

Detailed Comments (20)

1. 4.1 and 6.2: The rollout rule and displayed leads do not support parallel counterfactual trends

Issue: The rollout prioritizes high projected growth, while the event study displays statistically separated negative leads that trend toward zero before treatment and rising rather than stabilizing post-treatment coefficients.

Relevant text: The rollout prioritizes below-median productivity and above-median projected growth, while the results section states that Figure 2 shows no differential pre-trends and post-treatment stabilization.

Concern: The paper's central causal interpretation requires untreated productivity paths to be comparable; the selection rule and displayed dynamics point directly at differential counterfactual trends and anticipation or misspecified dynamics. A reader cannot interpret 0.034 as a causal average effect when treatment timing is selected on projected growth and the treated cohorts already follow different pre-treatment paths. The scoring rule directly selects on projected growth, and Figure 2 visibly shows negative pre-treatment coefficients converging toward zero; neither fact alone identifies the counterfactual, but together they contradict the paper's stated pre-trend interpretation.

Suggestions: Treat the current causal magnitude as provisional, report a design that addresses the forecast-based rollout rule and the displayed differential leads, and narrow the claim if that design does not support a stable effect. Show cohort-time estimates using not-yet-treated or never-treated controls, report conditional event-time leads using the published score and engineering variables, and give a pre-trend sensitivity or partial-identification analysis. If the leads persist, recast the result as descriptive or narrow it to the conditional comparison supported.

2. Section 5: The staggered TWFE coefficient has no defined cohort-time aggregation

Issue: Dynamic treatment effects under 2011-2018 staggered adoption make standard TWFE a weighted mixture, but the paper does not disclose that mixture and Figure 3 covers only five cohorts with a different comparison rule.

Relevant text:

Our baseline specification is the two-way fixed-effects regression

Concern: The sign and magnitude of a TWFE coefficient can depend on already-treated comparisons and implicit horizon weights, so positive cohort-specific bars do not prove that negative weighting is absent. The reported 3.4 percent cannot be mapped to an average treatment effect over a stated population and horizon. The manuscript uses standard two-way fixed effects under staggered adoption and displays dynamic effects; positive cohort bars for only 2011-2015 do not reveal the weights or comparisons in the baseline coefficient.

Suggestions: Replace or benchmark TWFE with a group-time or interaction-weighted estimator and define the aggregation over cohorts and horizons. Use a modern staggered-adoption estimator with not-yet-treated or never-treated controls, report cohort-time support, and show the aggregation weights and horizon. Retain TWFE only as a benchmark.

3. 2.2 and Appendix A: The empirical equation does not follow from the displayed production equation

Issue: Moving log labor to the left is algebraically inconsistent with its $1-\eta$ coefficient, and linearizing the model does not create the regression's standalone beta times broadband term.

Relevant text: The output equation contains $(1-\eta)$ times log labor, while the claimed estimating equation subtracts the full log-labor term and adds a standalone broadband coefficient.

Concern: The model is advertised as disciplining the estimating equation and mechanism, so an incorrect bridge undermines the structural meaning attached to the main and interaction coefficients. The regression's dependent variable, broadband main effect, and parameter interpretations appear structurally derived even though the displayed steps do not produce them. The displayed log-output equation has labor coefficient $1-\eta$; subtracting full log labor leaves $-\eta$ times log labor, and the stated linearization generates only an interaction term, not a standalone broadband term.

Suggestions: Correct the algebra and remove structural interpretations of beta and gamma that the corrected derivation does not deliver. Either derive a labor-productivity equation with the necessary labor term and explicit centering, or present the regression as a reduced-form specification and limit the model to a qualitative heterogeneity prediction.

4. Section 2.1: Proposition 1 is strict only in the switching regime used by the proof

Issue: The proposition states $G'(\theta) > 0$ for all θ , but the proof begins by imposing the switching regime and even displays only a weak inequality.

Relevant text: Proposition 1 states strict increase for all communication intensities, while the proof explicitly considers only the regime in which the firm switches from in-house production to remote procurement.

Concern: When both sourcing indicators are equal, $G(\theta) = 0$ and $G'(\theta) = 0$, so the model does not predict strict heterogeneity globally. The global comparative static is stronger than the model supports and is used to motivate the heterogeneity test. The proof establishes monotonicity conditional on one firm switching sourcing mode; outside that regime the two indicator terms coincide and the gain is constant at zero.

Suggestions: Restrict the proposition to the switching regime and use strict positivity in the derivative, or state the global result as weak monotonicity with explicit flat regimes. State strict increase conditional on the switching inequalities and flat gains otherwise, or state weak monotonicity globally; propagate the qualification to the empirical prediction.

5. Section 7: The policy section converts a level estimate into unsupported TFP, growth, and fiscal-payback objects

Issue: A 0.034 log labor-productivity level coefficient is relabeled as TFP and 3.4 percentage points of annual growth, then converted to an untraceable 410 million euros and treated as fiscal repayment.

Relevant text: The section moves from a 3.4 percent average-firm TFP gain to 3.4 percentage points of annual productivity growth, 410 million euros per year, six-year public repayment, and a cost-effectiveness ranking.

Concern: These are different economic objects; the transformations determine the headline policy return and the claim that broadband is among the most cost-effective policies. A reader could mistake a conditional labor-productivity coefficient for annual TFP growth and a gross output benefit for government revenue. The cost-to-claimed-benefit division is arithmetically about 5.61 years, but the manuscript does not establish the labor-productivity-to-TFP, level-to-annual-growth, sample-to-aggregate-euros, or gross-benefit-to-fiscal-repayment conversions.

Suggestions: Remove the TFP, annual-growth, repayment, and cost-effectiveness claims unless each conversion is rebuilt from traceable aggregates and an explicit welfare or fiscal framework. The minimum repair is to remove the TFP, annual-growth, euro, repayment, and comparative cost-effectiveness claims. A broader path would document the aggregate base, treatment dynamics, counterfactual, uncertainty, discounting, resource costs, and fiscal incidence.

6. Section 6.3: Communication-intensity heterogeneity does not identify task outsourcing

Issue: The 0.032 interaction with baseline communication intensity is consistent with several communication-related channels and does not by itself validate task outsourcing.

Relevant text:

Proposition 1 predicts that gains rise with communication intensity. Table 4 confirms this

Concern: The mechanism is a stated contribution and the formal model is used to interpret the heterogeneity coefficient, but neither the outcome nor the design observes sourcing behavior or rules out alternative channels. The paper presents one heterogeneous treatment effect as direct evidence for a specific organizational channel that was not measured. Table 4 supports a positive treatment-by-baseline-trait coefficient in the reported specification; it contains no task-sourcing outcome or comparison that separates outsourcing from other communication-intensive channels.

Suggestions: Describe the interaction as mechanism-consistent heterogeneity, not validation or identification of task outsourcing. Replace validates and mechanism we identify with language such as consistent with a communication-intensive channel. Retain a stronger claim only with a direct sourcing outcome and a discriminator against plausible alternatives.

7. Introduction: The broad priority claim is contradicted by close pre-existing firm-level studies

Issue: The manuscript claims the first causal evidence and says small firms were largely bypassed, despite prior causal firm-level broadband studies and a close staged municipal rollout DiD with firm productivity outcomes.

Relevant text: The paper claims the first causal evidence on broadband expansion and small-firm productivity and says existing work largely bypassed small firms.

Concern: At a leading field journal, contribution must be defined relative to the closest designs and outcomes rather than a selective aggregate/worker benchmark. The claimed novelty is broader than the literature record and obscures the paper's genuinely narrower contribution. The completed frontier search identifies prior causal firm-productivity studies and a particularly close staged municipal broadband DiD; the paper may still contribute nationwide under-50 coverage and communication-intensity heterogeneity.

Suggestions: Withdraw the first-causal-evidence and largely-bypassed claims, cite the closest work, and position the contribution on the dimensions that remain distinctive. Compare the verified firm-level studies directly on rollout design, sample, outcome, and heterogeneity, then use the same narrower framing in the abstract, introduction, and conclusion.

8. Section 4.2: Table 1 reports an employee mean above the employee maximum

Issue: The employee mean is 23.40 even though the employee maximum is 19.00 in the same table and sample.

Relevant text: Table 1 labels the employee mean as 23.40 and the employee maximum as 19.00.

Concern: An impossible mean-max relationship indicates a transcription, column-alignment, sample, or data-generation error that could extend beyond this cell. The sample description and basic data audit are not trustworthy until an impossible summary statistic is reconciled. The rendered table unambiguously displays a mean of 23.40 and a maximum of 19.00 for employees; the review cannot determine which number or column is wrong without the underlying data.

Suggestions: Reproduce Table 1 from the analysis sample, correct the row or column, and audit all displayed summary statistics against the generating output. Create the table directly from the final analysis sample, compare every cell with the generating output, and update the prose mean and sample description from that same output.

9. Section 4.2: The balanced panel conditions on post-treatment survival and positive sales

Issue: Firms must have positive sales throughout and a balanced 2008-2019 record, so inclusion depends on post-treatment outcomes that broadband could itself affect.

Relevant text: The sample requires at least four consecutive years and positive sales throughout, yet is described as a balanced panel of 9,400 firms observed in every year from 2008 to 2019.

Concern: If broadband changes survival, reporting, or positive-sales status, conditioning on those outcomes changes the population and can bias the productivity comparison among survivors. The headline effect is presented for small firms generally even though the analysis conditions on remaining observable with positive sales after treatment. The text requires positive sales throughout and then uses firms observed for all 12 years; this defines a survivor sample, but the direction and size of resulting selection cannot be learned from the supplied paper alone.

Suggestions: State the survivor estimand, show the full sample flow, and assess whether treatment affects inclusion, exit, or positive-sales status. Show counts at each restriction, compare treatment timing with exit and missingness, re-estimate on the broadest defensible unbalanced panel, and use bounds or a survivor-only claim if selection remains.

10. 5 and Appendix B: Municipality-level assignment is not reflected in inference for the dynamic and heterogeneity results

Issue: Treatment timing is common within municipalities, yet the event-study, placebo, and heterogeneity results use firm-clustered standard errors and the municipality-clustered check is limited to one static coefficient.

Relevant text: Broadband status is assigned from headquarters-municipality backbone service, while the empirical strategy says standard errors are clustered at the firm level throughout.

Concern: Firm clustering permits serial correlation within firms but not shocks shared by firms assigned together in the same municipality, so the precision of the most important supporting results may be overstated. A reader sees assignment at the municipality-year level but firm-clustered uncertainty for the results used to establish dynamics, the placebo, and the mechanism. Municipality clustering raises the static standard error only from 0.002 to 0.003 and preserves significance, but that alternative is not reported for the event study, placebo, or communication-intensity interaction.

Suggestions: Use municipality-level inference as the primary specification for every headline result and add a small-cluster sensitivity appropriate to 284 assignment units. Report municipality-clustered confidence intervals for all headline specifications and a wild-cluster bootstrap or randomization-based sensitivity where appropriate; revise significance language if conclusions change.

11. Section 8: The developing-economy extrapolation and higher-return claim are unsupported

Issue: The conclusion says the estimates generalize to developing economies and that returns are higher there without evidence on transportability or a mapped friction-response relationship.

Relevant text: The conclusion attributes universal communication-cost logic to the mechanism, generalizes the estimates to developing economies, and states that implied returns are correspondingly higher.

Concern: A mechanism being conceptually general does not make an effect magnitude portable, especially when take-up, infrastructure quality, firm composition, and complementary inputs differ. The conclusion invites a cross-country policy inference for a population and return schedule that the paper does not study. The simulated advanced-economy setting contains no transport design, developing-economy sample, or elasticity that maps larger communication frictions into correspondingly higher returns.

Suggestions: Remove the generalization and higher-return sentence, or recast it as an explicit hypothesis for future work. Delete the generalization and higher-return assertion or label it explicitly as a hypothesis whose relevance depends on take-up, complementary inputs, and firm composition.

12. 4.1 and Appendix B: Backbone arrival measures municipal availability rather than immediate firm access

Issue: The treatment indicator switches on at municipal backbone service, while firm connections and take-up occur later and are not observed at the firm level.

Relevant text: Treatment is the year a municipality's backbone enters commercial service; local firms connect within months and take-up exceeds 90 percent within two years.

Concern: This timing difference changes the estimand and can explain gradual post-treatment dynamics; it also rules out a literal immediate-access interpretation. The reader may interpret the estimate as the effect of firms receiving broadband at event time zero, although only municipal backbone service is observed. Commercial providers connect firms within months and take-up exceeds 90 percent only within two years; without firm connection dates, the coefficient is an intention-to-treat effect of municipal availability.

Suggestions: Define treatment as backbone availability and stop using immediate firm-access or take-up language for the coefficient. Use municipal backbone availability or program arrival throughout, describe the estimate as intention-to-treat, and interpret post-treatment dynamics as combining connection and response lags.

13. Section 6.2: The rounded placebo values do not support a precisely estimated zero claim

Issue: The paper calls -0.004 (0.002) statistically indistinguishable from zero and uses it to confirm a true effect, but the rounded values imply a t -statistic of -2.0 .

Relevant text: The text calls the placebo statistically indistinguishable from zero and says it confirms a true program effect; Table 3 reports -0.004 with standard error 0.002 and firm clustering.

Concern: A diagnostic with its own possible signal cannot be treated as proof of the identifying assumptions, and readers need the actual permutation distribution to interpret it. The placebo is described as zero and as confirming the design, although the displayed precision is borderline and the permutation procedure is undocumented. Using only the displayed rounded coefficient and standard error gives $t = -2.0$; exact p -values cannot be recovered, and one permutation coefficient without the randomization distribution cannot confirm a true program effect.

Suggestions: Report the exact placebo estimate, inference method, number and seed of permutations, randomization distribution, and use neutral language about what the exercise can test. Report exact coefficient, standard error or randomization p -value, permutation count and seed, and the distribution; replace confirms with a bounded statement about this diagnostic.

14. Abstract and results: The abstract reports 34 percent instead of 3.4 percent

Issue: The abstract's 34 percent magnitude conflicts with the 3.4 percent result in the body and conclusion.

Relevant text: The abstract reports 34 percent, whereas the introduction and conclusion report 3.4 percent and Table 2 reports 0.034 log points.

Concern: The abstract is the primary source for readers and indexing services, so this decimal error materially misstates the headline result. The paper's most visible summary overstates the reported coefficient by a factor of ten. The abstract says 34 percent; the introduction, Table 2, and conclusion consistently report 3.4 percent or 0.034 log points.

Suggestions: Change 34 percent to 3.4 percent and audit every headline magnitude for decimal consistency. Replace 34 percent with 3.4 percent and synchronize the chosen log-point or percent transformation across all summaries.

15. Section 6.1: The region-trend coefficient is 0.030 in Table 2, not 0.024

Issue: The results text gives 0.024 for the region-trend specification, while Table 2 column 3 displays 0.030.

Relevant text: The prose reports 0.024 for the region-trend coefficient, while Table 2 column 3 displays 0.030.

Concern: The region-trend check is a principal defense of identification, so its magnitude must be exact and synchronized. The reader cannot tell which region-trend estimate the paper intends to report. The prose and rendered table conflict; the review cannot determine whether the table or sentence reflects the intended specification without output.

Suggestions: Regenerate the table and prose from one estimation output and report the correct coefficient. Choose the final specification, regenerate Table 2, and update the sentence directly from that output.

16. Section 6.4: Table 5's labels and note describe incompatible specifications

Issue: Table 5 calls column 1 Baseline and column 2 No controls but its note says all columns include the full control set, while the text says column 1 repeats region trends.

Relevant text: The prose says column 1 repeats region trends and column 2 removes controls; Table 5 labels them Baseline and No controls but states that all columns contain the full control set.

Concern: The table is intended to show stability across specifications, but contradictory labels prevent readers from knowing what changed. The robustness estimates cannot be assigned to a known control and trend specification. The prose calls column 1 the region-trends specification and column 2 no controls, while the table labels column 1 Baseline and says every column has the full control set.

Suggestions: Correct the column labels and note from the actual specification matrix. Add explicit rows for firm controls, region trends, and cohort exclusions, and make each column heading and prose description match those rows.

17. Section 6.4: Figure 4 does not show within-sector broadband effects

Issue: The text says Figure 4 shows broadband effects within each sector, but the figure only reports sectoral composition and productivity levels.

Relevant text: The prose says the broadband effect is present within three sectors, while the caption identifies Figure 4 only as sectoral composition of the estimation sample.

Concern: Descriptive sector differences cannot establish within-sector treatment effects or robustness to sector composition. The prose attributes an effect result to a descriptive composition figure. Figure 4 contains sectoral employment and productivity levels only; no treatment coefficient, event-time contrast, or confidence interval is plotted.

Suggestions: Delete the within-sector effect sentence or replace Figure 4 with sector-specific treatment estimates and uncertainty. Either remove the effect sentence and describe Figure 4 as composition, or plot sector-specific estimates with comparable specifications and assignment-level intervals.

18. Section 4.2: The replication statement provides no traceable files or link

Issue: The sentence Replication files are available online has no accompanying repository identifier, link, or supplied package.

Relevant text: Replication files are available online.

Concern: The causal, table, figure, and policy calculations cannot be statically traced or executed from the materials provided for review. A reader is told that replication is available but has no way to locate or assess it. The paper states that files are online, but the supplied manuscript, bibliography, and review package contain no URL, DOI, repository identifier, code, or data.

Suggestions: Add a persistent repository link and a complete code-data documentation package, or remove the availability claim until the archive exists. Add a persistent repository identifier with code, documented data access, environment, seeds, and one-command table and figure reproduction; otherwise say files will be deposited rather than already available.

19. References: The Czernich et al. reference has the wrong journal and year

Issue: The bibliography lists Czernich et al. as an American Economic Review article from 2013, but the official record is a 2011 Economic Journal article.

Relevant text: The bibliography gives American Economic Review, 103(4):505-532, 2013 for Broadband Infrastructure and Economic Growth.

Concern: Accurate metadata is necessary for a reader to verify the cited antecedent and the literature comparison. The reference cannot be reliably located from the manuscript metadata. Official journal metadata establishes the bibliographic correction; it does not alter the source's country-level growth result.

Suggestions: Correct the record to The Economic Journal, 2011, 121(552):505-532, with the verified DOI. Replace the journal, year, volume, and issue with the official record and add the verified DOI.

20. Introduction: The cited Bloom et al. record does not verify the small-firm adoption claim

Issue: Bloom et al. is cited for delayed and incomplete small-firm technology adoption, yet the official abstract addresses US multinationals and IT productivity rather than that size comparison.

Relevant text: The introduction says small firms adopt new technologies later and less completely than large firms and cites Syverson and Bloom et al.

Concern: The motivation should rest on a source that directly studies adoption differences by firm size. A load-bearing motivation sentence appears sourced, but one inspected citation does not establish the stated size comparison. The official abstract verifies the paper and its multinational IT-productivity focus, but not the manuscript's statement that small firms adopt new technologies later and less completely than large firms.

Suggestions: Add a direct source for the size-by-adoption proposition or remove Bloom et al. from support for that sentence. Add a source whose inspected text establishes the sentence, or rewrite the motivation to the proposition actually supported by the cited papers.

Editing comments

Editing assessment

The manuscript is compact, well organized, and generally readable. Five objective copyediting or rendering defects remain: two prose errors, one clipped table heading, one broken figure label, and two malformed bibliography author names. The larger claim and exhibit-correspondence problems are routed to the substance report.

Paper-specific lens: Mixed causal-empirical paper with a stylized theory section, administrative panel design, graphical diagnostics, heterogeneity analysis, and policy calculation.

Strengths to preserve

- The paper’s question, treatment setting, outcome, and headline empirical object are introduced quickly.
- Section headings and exhibit numbering make the short manuscript easy to navigate.
- Most results paragraphs identify the relevant table or figure before interpreting it.

Highest-return editing revisions

1. **Restore the clipped Table 2 column heading.** Widen or rewrap the heading so Municipality cluster is fully visible. Shorten to Municipality-clustered SE or use a line break and verify the final PDF.
2. **Repair the broken productivity label in Figure 4.** Regenerate the figure with an embedded font and an intact Productivity label. Correct the plotting label, export with embedded fonts, and inspect the final PDF at normal and zoomed scale.
3. **Correct the malformed Callaway-Sant’Anna and Cameron-Miller reference names.** Correct the BibTeX author fields and verify the rendered names. Encode Pedro H. C. Sant’Anna and A. Colin Cameron in BibTeX’s family/given format and rebuild the bibliography.
4. **Remove the duplicated article in the data-source sentence.** Delete one occurrence of the. Change the the to the.
5. **Correct subject-verb agreement in the robustness summary.** Replace suggest with suggests. Use The set of estimates suggests, or rewrite as The estimates suggest.

Section-by-section reader audit

Title and abstract

Current job. State the question, design, main answer, mechanism interpretation, and policy relevance.

What works. The question and staggered-rollout setting are immediately clear.

Reader friction. The main friction is substantive rather than stylistic: the headline magnitude and policy scope are inconsistent with the body.

Revision direction. After resolving the substance findings, keep the abstract to one consistent magnitude, one bounded mechanism statement, and one defensible contribution.

Related comments. —

Model

Current job. Provide a compact economic mechanism and connect it to the heterogeneity regression.

What works. Primitives, sourcing choice, and the intended comparative static appear in a short sequence.

Reader friction. The prose itself is readable; the unresolved domain and algebra problems are scientific rather than editorial.

Revision direction. Preserve the compact presentation after correcting the proposition and derivation in the substance revision.

Related comments. —

Program and data

Current job. Explain rollout assignment, exposure timing, sample construction, and variables.

What works. The institutional chronology and administrative sources are easy to follow.

Reader friction. A duplicated article interrupts the opening data-source sentence.

Revision direction. Remove the duplicate word while retaining the concise data description.

Related comments. Comment 1: Remove the duplicated article in the data-source sentence

Results and robustness

Current job. Present the baseline, dynamic, mechanism, and robustness evidence through Tables 2-5 and Figures 2-4.

What works. The sequence from baseline to dynamics, heterogeneity, and robustness is intuitive.

Reader friction. One agreement error, the clipped Table 2 heading, and the broken Figure 4 label reduce finish and exhibit readability.

Revision direction. Correct the sentence and verify every exhibit label in the final PDF after the substantive specifications and captions are repaired.

Related comments. Comment 2: Correct subject-verb agreement in the robustness summary; Comment 3: Restore the clipped Table 2 column heading; Comment 4: Repair the broken productivity label in Figure 4

References

Current job. Make every cited antecedent and method source traceable.

What works. The list is short and most entries are visually recognizable.

Reader friction. Two central methods references have corrupted author-name rendering.

Revision direction. Correct the BibTeX author fields and visually inspect the rebuilt reference list.

Related comments. Comment 5: Correct the malformed Callaway-Sant’Anna and Cameron-Miller reference names

Terminology, definitions, and notation

Program names, small-firm terminology, section labels, and most notation are consistent. Load-bearing differences among labor productivity, TFP, growth, backbone availability, and firm access are substantive interpretation issues and are therefore handled in the referee report rather than as copyedits.

Program name

Status. consistent

Preferred form. National Broadband Program (NBP) at first use, then NBP

Variants checked. National Broadband Program; NBP

Locations checked. Abstract, introduction, institutional setting, and conclusion

Related comments. —

Small-firm label

Status. consistent

Preferred form. small firms as noun; small-firm as compound modifier

Variants checked. small firms; small-firm productivity; small-firm segment

Locations checked. Title, abstract, introduction, results, policy, and conclusion

Related comments. —

Exhibit naming**Status.** consistent**Preferred form.** Capitalized numbered Table and Figure; lowercase column**Variants checked.** Table; Figure; column**Locations checked.** Complete manuscript and all exhibit references**Related comments.** —**Tables and figures as writing**

The five tables and four figures are easy to locate. Table 2 has a clipped column heading and Figure 4 has a broken Productivity label; scientifically misleading table labels and figure-to-claim correspondence are routed to the substance report.

Mechanics and copyedit inventory**1. Spelling typo**

- **PDF p. 4:** Replace “We draw on the the Nordavian Business Register, which links audited annual accounts to” with “We draw on the Nordavian Business Register, which links audited annual accounts to”

See Comment 1: Remove the duplicated article in the data-source sentence for the full explanation and revision request.

Why it matters: The duplicated article interrupts the first data-source sentence.

Visible in both source and final render.

2. Agreement

- **PDF p. 8:** Replace “suggest that the baseline effect is stable across reasonable perturbations of the specification.” with “suggests that the baseline effect is stable across reasonable perturbations of the specification.”

See Comment 2: Correct subject-verb agreement in the robustness summary for the full explanation and revision request.

Why it matters: The agreement error is visible in a results-summary sentence.

The singular head noun is set.

3. Other

- **PDF p. 7:** Replace “(4) Municipality cluste” with “(4) Municipality-clustered SE”

See Comment 3: Restore the clipped Table 2 column heading for the full explanation and revision request.

Why it matters: The truncated heading hides the specification change in the alternative-inference column.

The crop and full page both show the clipping.

4. Spelling typo

- **PDF p. 10:** Replace “(b) Pro ductivity by sector” with “(b) Productivity by sector”

See Comment 4: Repair the broken productivity label in Figure 4 for the full explanation and revision request.

Why it matters: The main-figure panel title looks corrupted.

The defect is present in the main and standalone renders.

5. Other

- **PDF p. 10:** Replace “Brantly Callaway and C. Sant’Anna, Pedro H.’ Difference-in-differences with multiple time pe-” with “Brantly Callaway and Pedro H. C. Sant’Anna. Difference-in-differences with multiple time periods.”
- **PDF p. 10:** Replace “A.~Colin Cameron and Douglas L. Miller. A practitioner’s guide to cluster-robust inference.” with “A. Colin Cameron and Douglas L. Miller. A practitioner’s guide to cluster-robust inference.”

See Comment 5: Correct the malformed Callaway-Sant’Anna and Cameron-Miller reference names for the full explanation and revision request.

Why it matters: Two central methods references have visibly corrupted author names.

Both malformed names originate in the BibTeX author fields and survive into the render.

6. Article usage

No correction is needed in this category.

Why it matters: No systematic article-usage problem remains beyond the duplicate word recorded separately.

Countability, first mention, unique institutional objects, and generic categories were checked sentence by sentence.

7. Punctuation

No correction is needed in this category.

Why it matters: Punctuation is generally controlled and does not impede interpretation.

Commas, colons, semicolons, possessives, dashes, parentheses, and equation punctuation were checked.

8. Tense voice

No correction is needed in this category.

Why it matters: Tense and voice are stable enough for a short empirical paper.

Present-tense interpretation, past-tense setting description, active/passive voice, and author perspective were checked.

Style and writing improvements

No additional style revision is recommended beyond the prioritized items above.

Redundancy and repetition

No costly repetition or conflicting duplicate framing was identified.

Detailed Editing Comments (5)

1. Manuscript: Remove the duplicated article in the data-source sentence

Issue: The sentence begins We draw on the the Nordavian Business Register.

Relevant text:

We draw on the the Nordavian Business Register, which links audited annual accounts to

Concern: This is an objective copyediting error in a prominent data description. The duplicated word interrupts the sample-description sentence. The duplication is visible in the rendered paragraph.

Suggestions: Delete one occurrence of the. Change the the to the.

2. Manuscript: Correct subject-verb agreement in the robustness summary

Issue: The singular subject The set of estimates is paired with suggest.

Relevant text:

suggest that the baseline effect is stable across reasonable perturbations of the specification.

Concern: This is an objective agreement error. The agreement error distracts in a results-summary sentence. The singular subject set takes suggests.

Suggestions: Replace suggest with suggests. Use The set of estimates suggests, or rewrite as The estimates suggest.

3. Section 6.1: Restore the clipped Table 2 column heading

Issue: Table 2 column 4 visibly truncates Municipality cluster.

Relevant text: The visible heading reads (4) Municipality cluste and is clipped at the table's right edge.

Concern: The clipped label obscures the specification change that the column is meant to document. The principal alternative-inference column has an incomplete label. The rendered heading ends at Municipality cluste and is not a text-extraction artifact.

Suggestions: Widen or rewrap the heading so Municipality cluster is fully visible. Shorten to Municipality-clustered SE or use a line break and verify the final PDF.

4. Figure 4: Repair the broken productivity label in Figure 4

Issue: Figure 4 panel (b) renders Productivity with a visible break or missing glyph.

Relevant text: The panel title visibly renders Pro ductivity rather than Productivity.

Concern: A final-presentation defect in a main figure reduces legibility and signals an unverified graphics export. The panel title looks corrupted in the final PDF. The rendered panel title contains a visible spacing or missing-glyph break in Productivity.

Suggestions: Regenerate the figure with an embedded font and an intact Productivity label. Correct the plotting label, export with embedded fonts, and inspect the final PDF at normal and zoomed scale.

5. References: Correct the malformed Callaway-Sant’Anna and Cameron-Miller reference names

Issue: The bibliography renders C. Sant’Anna, Pedro H. after Callaway and A.Œolin Cameron instead of clean author names.

Relevant text: The rendered references show Brantly Callaway and C. Sant’Anna, Pedro H. and A.Œolin Cameron and Douglas L. Miller.

Concern: The defects impede citation recognition and are objective bibliography-formatting errors. Readers encounter visibly corrupted author names in two central methods references. The rendered bibliography reverses and corrupts author-name fields for two references; the paper titles and publication details remain recognizable.

Suggestions: Correct the BibTeX author fields and verify the rendered names. Encode Pedro H. C. Sant’Anna and A. Colin Cameron in BibTeX’s family/given format and rebuild the bibliography.

Revision plan

Objective: improve the paper for leading-field by resolving the verified concerns below.

How to use this plan

Work from P0 to P2. A later review confirms whether each revision resolves the concern by comparing it with the **Done when** condition.

Unless an item says otherwise, the current design can support the stated repair. Items that need new evidence or claim narrowing say so explicitly.

P0 — essential before submission

Comment 1: The rollout rule and displayed leads do not support parallel counterfactual trends

- **Severity:** critical
- **Action:** Re-estimate the causal design around the actual rollout score and displayed pre-treatment dynamics. Show cohort-time estimates using not-yet-treated or never-treated controls, report conditional event-time leads using the published score and engineering variables, and give a pre-trend sensitivity or partial-identification analysis. If the leads persist, recast the result as descriptive or narrow it to the conditional comparison supported.
- **Payoff:** This repair determines whether the paper’s headline causal result is credible at a leading field journal.
- **Done when:** The revised paper either presents a transparent design in which the scoring-rule selection and differential leads no longer overturn a clearly defined effect, or removes the causal-average and permanent-effect claims throughout.
- **Effort:** weeks
- **Dependencies:** None

P1 — substantive revision

Comment 2: The empirical equation does not follow from the displayed production equation

- **Severity:** major
- **Action:** Rewrite the model-to-regression bridge from the correctly transformed outcome. Either derive a labor-productivity equation with the necessary labor term and explicit centering, or present the regression as a reduced-form specification and limit the model to a qualitative heterogeneity prediction.
- **Payoff:** This removes a direct formal objection and clarifies what, if anything, the model identifies empirically.
- **Done when:** Every algebraic step is valid, beta and gamma have one consistent interpretation across text and appendix, and the empirical variable definitions match the derivation.
- **Effort:** days
- **Dependencies:** None

Comment 3: Proposition 1 is strict only in the switching regime used by the proof

- **Severity:** major
- **Action:** Align the proposition's domain and inequality with the proof. State strict increase conditional on the switching inequalities and flat gains otherwise, or state weak monotonicity globally; propagate the qualification to the empirical prediction.
- **Payoff:** A precise proposition preserves the useful qualitative intuition without claiming a false global result.
- **Done when:** The proposition, proof, and mechanism language share the same domain and strictness, with the non-switching cases handled explicitly.
- **Effort:** hours
- **Dependencies:** None

Comment 4: The broad priority claim is contradicted by close pre-existing firm-level studies

- **Severity:** major
- **Action:** Replace the broad priority claim with the narrower contribution established in the literature comparison above. Compare the verified firm-level studies directly on rollout design, sample, outcome, and heterogeneity, then use the same narrower framing in the abstract, introduction, and conclusion.
- **Payoff:** Accurate positioning makes the paper's real contribution visible and avoids an easily rebutted novelty claim.
- **Done when:** Every priority statement is qualified, the closest antecedents are cited and compared on design, outcome, sample, and result, and the abstract, introduction, and conclusion use the same defensible contribution.
- **Effort:** days
- **Dependencies:** None

Comment 5: Table 1 reports an employee mean above the employee maximum

- **Severity:** major
- **Action:** Regenerate and reconcile the summary-statistics table. Create the table directly from the final analysis sample, compare every cell with the generating output, and update the prose mean and sample description from that same output.
- **Payoff:** A clean data-summary audit is a prerequisite for confidence in the empirical package.
- **Done when:** No mean lies outside its displayed range and a reproducible table output matches every Table 1 cell and the adjacent text.
- **Effort:** days
- **Dependencies:** None

Comment 6: The balanced panel conditions on post-treatment survival and positive sales

- **Severity:** major
- **Action:** Make the sample estimand and selection consequences transparent. Show counts at each restriction, compare treatment timing with exit and missingness, re-estimate on the broadest defensible unbalanced panel, and use bounds or a survivor-only claim if selection remains.
- **Payoff:** This establishes which firms the estimate describes and prevents a hidden post-treatment conditioning objection.
- **Done when:** The paper defines its population as survivors or demonstrates that treatment-related inclusion does not drive the result, with a reproducible sample-flow table.
- **Effort:** days
- **Dependencies:** None

Comment 7: The developing-economy extrapolation and higher-return claim are unsupported

- **Severity:** minor
- **Action:** Keep the conclusion within the studied setting and estimand. Delete the generalization and higher-return assertion or label it explicitly as a hypothesis whose relevance depends on take-up, complementary inputs, and firm composition.
- **Payoff:** The conclusion becomes credible without sacrificing the paper's setting-specific contribution.
- **Done when:** No sentence presents developing-economy effect sizes or rankings as established by this study.
- **Effort:** hours
- **Dependencies:** None

Comment 8: Backbone arrival measures municipal availability rather than immediate firm access

- **Severity:** minor
- **Action:** Name the treatment and estimand consistently. Use municipal backbone availability or program arrival throughout, describe the estimate as intention-to-treat, and interpret post-treatment dynamics as combining connection and response lags.
- **Payoff:** A precise treatment definition makes the dynamic pattern and policy estimand easier to interpret.
- **Done when:** Treatment language, event-time labels, and conclusions no longer equate backbone service mechanically with firm-level access at arrival.
- **Effort:** hours
- **Dependencies:** None

Comment 9: The region-trend coefficient is 0.030 in Table 2, not 0.024

- **Severity:** minor
- **Action:** Synchronize the region-trend estimate. Choose the final specification, regenerate Table 2, and update the sentence directly from that output.
- **Payoff:** The correction removes uncertainty about a central robustness result.
- **Done when:** The text, table, and reproducible output report the same coefficient and specification.
- **Effort:** hours
- **Dependencies:** None

Comment 10: Table 5's labels and note describe incompatible specifications

- **Severity:** minor
- **Action:** Rebuild Table 5's specification labels and note. Add explicit rows for firm controls, region trends, and cohort exclusions, and make each column heading and prose description match those rows.
- **Payoff:** A self-contained robustness table lets readers evaluate the intended perturbations.
- **Done when:** Every Table 5 column has an unambiguous specification and agrees with the surrounding text and estimation output.
- **Effort:** hours
- **Dependencies:** None

Comment 11: Figure 4 does not show within-sector broadband effects

- **Severity:** minor
- **Action:** Make the figure and associated claim correspond. Either remove the effect sentence and describe Figure 4 as composition, or plot sector-specific estimates with comparable specifications and assignment-level intervals.
- **Payoff:** The correction prevents descriptive evidence from being mistaken for a robustness test.
- **Done when:** The prose describes exactly what Figure 4 displays, and any retained within-sector effect claim has direct exhibit support.
- **Effort:** hours
- **Dependencies:** None

Comment 12: The Czernich et al. reference has the wrong journal and year

- **Severity:** minor
- **Action:** Correct the bibliographic fields. Replace the journal, year, volume, and issue with the official record and add the verified DOI.
- **Payoff:** The correction restores traceability of a prominent antecedent.
- **Done when:** The rendered reference and BibTeX match the official Economic Journal record.
- **Effort:** hours
- **Dependencies:** None

Comment 13: The cited Bloom et al. record does not verify the small-firm adoption claim

- **Severity:** minor
- **Action:** Use a direct firm-size adoption citation. Add a source whose inspected text establishes the sentence, or rewrite the motivation to the proposition actually supported by the cited papers.
- **Payoff:** Direct source support removes a preventable literature-verification distraction.
- **Done when:** The sentence has a page-verifiable source for its firm-size comparison or no longer makes that comparison.
- **Effort:** hours
- **Dependencies:** None

Comment 14: The staggered TWFE coefficient has no defined cohort-time aggregation

- **Severity:** major
- **Action:** Define and estimate the treatment-effect aggregation that corresponds to the headline magnitude. Use a modern staggered-adoption estimator with not-yet-treated or never-treated controls, report cohort-time support, and show the aggregation weights and horizon. Retain TWFE only as a benchmark.
- **Payoff:** A transparent estimand is necessary to interpret the headline magnitude and compare it with the literature.
- **Done when:** A reader can reproduce the headline average from displayed cohort-time effects and weights, and the causal interpretation is limited to the cohorts and horizons with support.
- **Effort:** days
- **Dependencies:** Comment 1: The rollout rule and displayed leads do not support parallel counterfactual trends

Comment 15: Communication-intensity heterogeneity does not identify task outsourcing

- **Severity:** major
- **Action:** Calibrate the mechanism claim to the evidence actually observed. Replace validates and mechanism we identify with language such as consistent with a communication-intensive channel. Retain a stronger claim only with a direct sourcing outcome and a discriminator against plausible alternatives.
- **Payoff:** The narrower wording preserves an informative heterogeneity result without overselling the channel.
- **Done when:** The abstract, model discussion, heterogeneity section, and conclusion consistently distinguish predicted heterogeneity from identified task outsourcing, or new direct mechanism evidence supports the stronger statement.
- **Effort:** hours
- **Dependencies:** Comment 2: The empirical equation does not follow from the displayed production equation; Comment 3: Proposition 1 is strict only in the switching regime used by the proof

Comment 16: The replication statement provides no traceable files or link

- **Severity:** minor
- **Action:** Make the replication claim traceable. Add a persistent repository identifier with code, documented data access, environment, seeds, and one-command table and figure reproduction; otherwise say files will be deposited rather than already available.
- **Payoff:** A traceable package is important for a design whose key diagnostics and table labels require reconciliation.
- **Done when:** The manuscript points to an accessible archive whose contents reproduce the analysis artifacts or accurately discloses the archive's availability boundary.
- **Effort:** days
- **Dependencies:** Comment 5: Table 1 reports an employee mean above the employee maximum; Comment 9: The region-trend coefficient is 0.030 in Table 2, not 0.024; Comment 10: Table 5's labels and note describe incompatible specifications

Comment 17: The policy section converts a level estimate into unsupported TFP, growth, and fiscal-payback objects

- **Severity:** major
- **Action:** Limit the policy section to interpretable, traceable implications of the estimated labor-productivity level effect. The minimum repair is to remove the TFP, annual-growth, euro, repayment, and comparative cost-effectiveness claims. A broader path would document the aggregate base, treatment dynamics, counterfactual, uncertainty, discounting, resource costs, and fiscal incidence.
- **Payoff:** This preserves the empirical result while preventing the policy conclusion from outrunning the measured object.
- **Done when:** Every retained policy number is reproducible from cited inputs with consistent units and horizon, and the prose distinguishes gross productivity benefits from welfare and fiscal repayment.
- **Effort:** days
- **Dependencies:** Comment 1: The rollout rule and displayed leads do not support parallel counterfactual trends; Comment 14: The staggered TWFE coefficient has no defined cohort-time aggregation

Comment 18: Municipality-level assignment is not reflected in inference for the dynamic and heterogeneity results

- **Severity:** major
- **Action:** Align the uncertainty calculation with municipality-level assignment throughout. Report municipality-clustered confidence intervals for all headline specifications and a wild-cluster bootstrap or randomization-based sensitivity where appropriate; revise significance language if conclusions change.
- **Payoff:** Consistent assignment-level inference makes the reported precision interpretable without discarding the reassuring static check.
- **Done when:** Every central estimate, Figure 2 interval, placebo, and heterogeneity result has municipality-level primary inference and an explicit count of assignment clusters.
- **Effort:** days
- **Dependencies:** Comment 14: The staggered TWFE coefficient has no defined cohort-time aggregation

Comment 19: The abstract reports 34 percent instead of 3.4 percent

- **Severity:** minor
- **Action:** Correct the abstract magnitude. Replace 34 percent with 3.4 percent and synchronize the chosen log-point or percent transformation across all summaries.
- **Payoff:** The correction prevents a factor-of-ten misstatement of the main result.
- **Done when:** Every headline location reports the same correctly transformed magnitude and unit.
- **Effort:** hours
- **Dependencies:** Comment 1: The rollout rule and displayed leads do not support parallel counterfactual trends; Comment 14: The staggered TWFE coefficient has no defined cohort-time aggregation

Comment 20: The rounded placebo values do not support a precisely estimated zero claim

- **Severity:** minor
- **Action:** Make the placebo reproducible and calibrate its evidentiary role. Report exact coefficient, standard error or randomization p-value, permutation count and seed, and the distribution; replace confirms with a bounded statement about this diagnostic.
- **Payoff:** Transparent diagnostic reporting prevents an avoidable challenge to the design discussion.
- **Done when:** The placebo can be reproduced and its wording matches the exact inference without claiming it proves the design.
- **Effort:** hours
- **Dependencies:** Comment 18: Municipality-level assignment is not reflected in inference for the dynamic and heterogeneity results

P2 – copyediting and optional polish

Comment 21: Remove the duplicated article in the data-source sentence

- **Severity:** minor
- **Action:** Delete the duplicate article. Change the the to the.
- **Payoff:** Required copyedit to remove a visible duplicated article in the data description.
- **Done when:** The rendered sentence contains one the before Nordavian Business Register.
- **Effort:** hours
- **Dependencies:** None

Comment 22: Correct subject-verb agreement in the robustness summary

- **Severity:** minor
- **Action:** Correct the verb. Use The set of estimates suggests, or rewrite as The estimates suggest.
- **Payoff:** Required copyedit to restore grammatical agreement in the results discussion.
- **Done when:** The rendered sentence has subject-verb agreement.
- **Effort:** hours
- **Dependencies:** None

Comment 23: Restore the clipped Table 2 column heading

- **Severity:** minor
- **Action:** Fit the full column heading within the table. Shorten to Municipality-clustered SE or use a line break and verify the final PDF.
- **Payoff:** Required exhibit polish so every Table 2 specification label is readable.
- **Done when:** The entire column heading is legible without clipping in the rendered PDF.
- **Effort:** hours
- **Dependencies:** None

Comment 24: Repair the broken productivity label in Figure 4

- **Severity:** minor
- **Action:** Regenerate the label and embed the font. Correct the plotting label, export with embedded fonts, and inspect the final PDF at normal and zoomed scale.
- **Payoff:** Required exhibit polish so Figure 4's productivity label renders correctly.
- **Done when:** Figure 4 panel (b) renders Productivity without a gap, box, or missing glyph.
- **Effort:** hours
- **Dependencies:** None

Comment 25: Correct the malformed Callaway-Sant'Anna and Cameron-Miller reference names

- **Severity:** minor
- **Action:** Correct the two author fields. Encode Pedro H. C. Sant'Anna and A. Colin Cameron in BibTeX's family/given format and rebuild the bibliography.
- **Payoff:** Required bibliography polish.
- **Done when:** Both references render the authors' names in standard order without stray punctuation or diacritic corruption.
- **Effort:** hours
- **Dependencies:** None